Consider the system

$$x_{k+1} = x_k + u_k + w_k, \qquad k = 0, 1, 2, 3,$$

with initial state $x_0 = 5$, and the cost function

$$\sum_{k=0}^{3} (x_k^2 + u_k^2).$$

Apply the DP algorithm for the following three cases:

(a) The control constraint set $U_k(x_k)$ is $\{u \mid 0 \le x_k + u \le 5,\ u:\ \text{integer}\}$ for all x_k and k, and the disturbance w_k is equal to zero for all k.

(b) The control constraint is as in part (a) and the disturbance w_k takes the values -1 and 1 with equal probability $1/2$ for all x_k and u_k, except if $x_k + u_k$ is equal to 0 or 5, in which case $w_k = 0$ with probability 1.

(a) $\because W_k = 0$

$\therefore X_{k+1} = X_k + u_k + W_k = X_k + u_k$, $k = 0, 1, 2, 3$

$\therefore$ 由 $0 \le X_k + u \le 5$ 得 $0 \le X_{k+1} \le 5$, $k = 0, 1, 2, 3$

又 $\because X_0 = 5$

$\therefore 0 \le X_k \le 5$, $k = 0, 1, 2, 3, 4$, 且 $X_k \in N$ ---------- ①

令 $J_4(X_4) = g_4(X_4) = 0$

① 当 $k=3$ 时, $J_3(X_3) = \min\limits_{u_3} E_{w_k} (X_3^2 + u_3^2 + 0) = \min\limits_{u_3} (X_3^2 + u_3^2) \geq X_3^2$ ---- ②

由①和 $\{u \mid 0 \le X_k + u \le 5\}$ 知 u_3 为 $\{-X_3, 1-X_3, \cdots, 5-X_3\}$

而 $X_3 \in [0, 5]$ 且 $X_3 \in N$, 故不论 X_3 为何值, u_3 均可取 0

$\therefore J_3(X_3) = X_3^2$, $\forall X_3$, $u_3 = 0$ ---------- ③

② 当 $k=2$ 时, $J_2(X_2) = \min\limits_{u_2} (X_2^2 + u_2^2 + J_3(X_2 + u_2))$

$$= \min_{u_2} (X_2^2 + u_2^2 + (X_2 + u_2)^2)$$

遍历 $X_2 = 0, 1, 2, 3, 4, 5$ 和 $u_2 = -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$ 可得.

X_2	0	1	2	3	4	5
u_2	0	-1, 0	-1	-2, -1	-2	-3, -2
$J_2(X_2)$	0	2	6	14	24	38

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③ 当k=1时，$J_1(x_1) = \min\limits_{u_1} (x_1^2 + u_1^2 + J_2(x_1+u_1))$，

同理，遍历 x_1 和 u_1 可得：

x_1	0	1	2	3	4	5
u_1	0	-1	-1	-2	-2	3
$J_1(x_1)$	0	2	7	15	26	40

⑤

④ 当k=0时，$J_0(x_0) = \min\limits_{u_0} (x_0^2 + u_0^2 + J_1(x_0+u_0))$

$\because x_0=5$，$0 \le x_0+u_0 \le 5$

$\therefore -5 \le u_0 \le 0$

$\therefore$ 当 $u_0=-3$ 时，$J_0(x_0)$ 有最小值41 ⑥

综上所述，$u=(u_0, u_1, u_2, u_3)$ 由③④⑤⑥所示。

(b) 由(a)及(b)题意可知，仍有 $x_k \in [0,5]$，$x_k \in N$。令 $J_4(x_4)=0$

① 当k=3时，$J_3(x_3) = \min\limits_{u_3} (x_3^2 + u_3^2)$，

由(a)②可知，$u_3(x_3)=0$，有最小值 $J_3(x_3) = x_3^2$ ①

② 当k=2时，$J_2(x_2) = \begin{cases} \min\limits_{u_2} (x_2^2 + u_2^2 + J_3(x_2+u_2)), & u_2+x_2=5 \text{ 或 } 0 \\ \min\limits_{u_2} (x_2^2 + u_2^2 + \frac{1}{2}J_3(x_2+u_2+1) + \frac{1}{2}J_3(x_2+u_2-1)), & \text{其他} \end{cases}$

遍历 x_2 与 u_2，得

x_2	0	1	2	3	4	5
u_2	0	-1	-1	-2/-1	-2	-2/-3
$J_2(x_2)$	0	2	7	15	21	39

②

③ 当k=1时，$J_1(x_1) = \begin{cases} \min (x_1^2 + u_1^2 + J_2(x_1+u_1)), & x_1+u_1=5 \text{ 或 } 0 \\ \min (x_1^2 + u_1^2 + \frac{1}{2}J_2(x_1+u_1+1) + \frac{1}{2}J_2(x_1+u_1-1)), & \text{其他} \end{cases}$

遍历 x_1 和 u_1 得

x_1	0	1	2	3	4	5
u_1	0	-1	-2	-2	-2/-3	-3
$J_1(x_1)$	0	2	8	16.5	28.5	42.5

…③

④ 当 $k=0$ 时，$J_0(x_0)=\min\limits_{u_0}\left(x_0^2+u_0^2+J_1(x_0+u_0+w_0)\right)$

由于 $x_0=5$，故 $-5\le u_0\le 0$，$J_0(x_0)=\begin{cases}\min\limits_{u_0}\left(25+u_0^2+J_1(5+u_0)\right), & 5+u_0=5\text{或}0\\ \min\limits_{u_0}\left(25+u_0^2+\frac{1}{2}J_1(5+u_0+1)+\frac{1}{2}J_1(5+u_0-1)\right), & \text{其他}\end{cases}$

遍历 u_0 可得 $u_0(x_0)=-3$ 时，$J_0(x_0)=43.25$ 为最小值，- - - - - - ④

综上所述，$u=(u_0,u_1,u_2,u_3)$ 由①②③④所示。